Strict Subadditivity for a Fourth-Order Cubic–Quintic Hamiltonian in Two Dimensions

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Abstract

We consider the mass-constrained infimum of a fourth-order Hamiltonian with a focusing quartic energy term and a defocusing sextic energy term on $H^2(\mathbb{R}^2;\mathbb{C})$. We establish strict subadditivity for every interior splitting of each mass $0 < m \le m_B$, where

$$m_B = \frac{3\beta}{\gamma\sqrt{k^4 + 3\beta^2/(32\gamma)}} \qquad (\beta, \gamma, k > 0).$$

The proof first constructs negative-energy trials at every positive mass, then uses an exact mass-preserving dilation to replace arbitrary minimizing sequences by sequences satisfying a scalar stationarity identity. A Fourier L^∞ estimate and a square-completion identity control the sextic-to-quartic ratio and yield a quantitative strict decrease of energy per unit mass. No attainment of the constrained infimum is assumed. An exact-arithmetic corollary verifies the sufficient condition for a specified frozen parameter set. The result is confined to this two-dimensional combined-power variational problem and does not assert a sharp critical mass or conclusions about physical confinement lifetimes.

Keywords. Fourth-order nonlinear Schrödinger energy; strict subadditivity; mass constraint; combined nonlinearities; concentration–compactness; mass-preserving dilation.
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1 Introduction and main result

Let

$$X = H^2(\mathbb{R}^2;\mathbb{C}), \qquad A = \Delta + k^2,$$

and, for $\beta, \gamma, k > 0$, define

$$H(u) = \|Au\|_2^2 - \frac{\beta}{2}\|u\|_4^4 + \frac{\gamma}{3}\|u\|_6^6. \tag{1.1}$$

For $m > 0$, let

$$I(m) = \inf\{H(u) : u \in X,\ \|u\|_2^2 = m\}. \tag{1.2}$$

All integrals are over $\mathbb{R}^2$ with Lebesgue measure. No radial, real-valued, periodic, band-limited, or other symmetry restriction is imposed.

For orientation, mixed-dispersion fourth-order nonlinear Schrödinger variational problems have been studied by Fernández, Jeanjean, Mandel, and Mariş [1], and combined-power mixed-dispersion problems by Ma, Chang, and Feng [2]. The result proved here is the explicit sufficient mass interval for strict subadditivity of the *particular* Hamiltonian (1.1). No assertion of an exhaustive literature-based priority or optimality is made.

Theorem 1.1 (Explicit strict-binding range)**.** *Let $\beta, \gamma, k > 0$ and set*

$$\rho = k^4 + \frac{3\beta^2}{32\gamma}, \qquad \sigma = \sqrt{\rho}, \qquad m_B = \frac{3\beta}{\gamma\sigma}. \tag{1.3}$$

For every $0 < m \le m_B$ and every $0 < \mu < m$,

$$\boxed{I(m) < I(\mu) + I(m-\mu).} \tag{1.4}$$

In particular, equality cannot occur at an interior splitting in this range. The statement does not require the existence of a minimizer for $I(s)$ at any mass s.

Our principal intermediate estimate, with $e(s) = I(s)/s$, is

$$e(a) - e(b) \ge \frac{(b-a)}{a^2}\left(1 - \frac{a+b}{2m_B}\right)[-I(a)] > 0 \quad \text{for } 0 < a < b,\ a + b < 2m_B. \tag{1.5}$$

This positive defect at the level of the infimum is more informative than a strict trial-function inequality, which need not remain strict after taking infima.

2 Basic estimates and negative constrained energy

For $u \in X$ introduce

$$M(u) = \|u\|_2^2, \quad D(u) = \|\Delta u\|_2^2, \quad K(u) = \|\nabla u\|_2^2,$$
$$T(u) = \|Au\|_2^2, \quad Q(u) = \|u\|_4^4, \quad S(u) = \|u\|_6^6.$$

Integration by parts, or Plancherel's theorem, gives

$$T = D - 2k^2K + k^4M \ge 0. \tag{2.1}$$

The embedding $H^2(\mathbb{R}^2) \hookrightarrow L^p(\mathbb{R}^2)$ for finite $p \ge 2$ shows that H is well-defined and continuous on X.

Lemma 2.1 (Coercive lower bound)**.** *For each $m > 0$,*

$$I(m) \ge -\frac{3\beta^2}{16\gamma}\, m > -\infty. \tag{2.2}$$

Furthermore, a fixed-mass sequence with energies bounded above is bounded in $H^2(\mathbb{R}^2)$.

Proof. For $r \ge 0$, square completion gives the *exact* identity

$$-\frac{\beta}{2}r^2 + \frac{\gamma}{3}r^3 = \frac{\gamma}{3}r\left(r - \frac{3\beta}{4\gamma}\right)^2 - \frac{3\beta^2}{16\gamma}r. \tag{2.3}$$

Integrating with $r = |u|^2$ yields

$$H(u) \ge T(u) - \frac{3\beta^2}{16\gamma}M(u),$$

which proves (2.2). If $M(u_n) = m$ and $H(u_n) \le C_0$, then $T(u_n) \le C_0 + 3\beta^2 m/(16\gamma)$. Since

$$\|\Delta u_n\|_2 \le \|Au_n\|_2 + k^2\sqrt{m}, \qquad K(u_n) \le \sqrt{M(u_n)D(u_n)},$$

the H^2 norms remain bounded. □

Lemma 2.2 (Strict negativity at every positive mass)**.** *For every $m > 0$, $I(m) < 0$.*

Proof. For $a, b > 0$, set

$$f_{a,b}(x,y) = \frac{1}{\sqrt{\pi ab}} \exp\left(-\frac{x^2}{2a^2} - \frac{y^2}{2b^2}\right), \qquad \|f_{a,b}\|_2 = 1,$$

and, for $R > 0$, consider the complex anisotropic trial

$$u_R(x,y) = \sqrt{m}\, R^{-1} f_{a,b}(x/R, y/R) e^{ikx}.$$

It satisfies $M(u_R) = m$. Differentiating the Gaussian and integrating shows that

$$H(u_R) = \frac{c_2}{R^2} + \frac{c_4}{R^4}, \tag{2.4}$$

where

$$c_2 = \frac{2mk^2}{a^2} - \frac{\beta m^2}{4\pi ab}, \tag{2.5}$$

$$c_4 = m\left(\frac{3}{4a^4} + \frac{1}{2a^2b^2} + \frac{3}{4b^4}\right) + \frac{\gamma m^3}{9\pi^2 a^2 b^2} > 0. \tag{2.6}$$

For any fixed $m > 0$, choose $a/b > 8\pi k^2/(\beta m)$, giving $c_2 < 0$. Then $R^2 = -2c_4/c_2$ is positive and

$$H(u_R) = -\frac{c_2^2}{4c_4} < 0.$$

As u_R has mass m, the constrained infimum is strictly negative. □

Combining the two lemmas,

$$-\frac{3\beta^2}{16\gamma} m \le I(m) < 0 \qquad \text{for all } m > 0. \tag{2.7}$$

3 Exact adjustment of minimizing sequences

For $\lambda > 0$, let

$$u_\lambda(x) = \lambda u(\lambda x). \tag{3.1}$$

In two dimensions, $M(u_\lambda) = M(u)$, while

$$D(u_\lambda) = \lambda^4 D(u), \qquad K(u_\lambda) = \lambda^2 K(u), \qquad Q(u_\lambda) = \lambda^2 Q(u), \qquad S(u_\lambda) = \lambda^4 S(u).$$

Thus

$$H(u_\lambda) = \lambda^4 \underbrace{\left(D + \frac{\gamma}{3} S\right)}_{P} - \lambda^2 \underbrace{\left(2k^2 K + \frac{\beta}{2} Q\right)}_{B} + k^4 M. \tag{3.2}$$

For nonzero $u \in X$, both $P > 0$ and $B > 0$.

Lemma 3.1 (Dilation-stationary minimizing sequences)**.** *For every $a > 0$ there exists a minimizing sequence $(w_n) \subset X$ such that*

$$M(w_n) = a, \qquad H(w_n) \longrightarrow I(a), \tag{3.3}$$

and each w_n satisfies the exact identity

$$\boxed{D(w_n) - k^2 K(w_n) - \frac{\beta}{4} Q(w_n) + \frac{\gamma}{3} S(w_n) = 0.} \tag{3.4}$$

No attainment assumption is needed.

Proof. For a nonzero u, write $z = \lambda^2$. Equation (3.2) is the strictly convex quadratic

$$Pz^2 - Bz + k^4 M,$$

whose unique positive minimum occurs at $z_0 = B/(2P)$. Defining $w = u_{\sqrt{z_0}}$ gives

$$H(u_\lambda) - H(w) = P(\lambda^2 - z_0)^2 \geq 0. \tag{3.5}$$

Stationarity at $\lambda = \sqrt{z_0}$ reads $2P(w) = B(w)$, which is precisely (3.4).

Start with any mass-a minimizing sequence (u_n). Apply this adjustment individually to obtain w_n. The dilation preserves mass and lowers or preserves energy, so

$$I(a) \leq H(w_n) \leq H(u_n) \longrightarrow I(a).$$

This proves (3.3) and (3.4). Boundedness in H^2 also follows from Lemma 2.1, although no weak or strong limit is required here. □

4 Sextic-to-quartic control

The stationarity identity produces a stronger bound than a direct application of an arbitrary-profile Sobolev estimate.

Lemma 4.1 (Square completion on the stationary dilation)**.** *If $w \in X \setminus \{0\}$ satisfies (3.4), then*

$$D(w) + T(w) + \frac{2\gamma}{3} \int_{\mathbb{R}^2} |w|^2 \left(|w|^2 - \frac{3\beta}{8\gamma} \right)^2 \mathrm{d}x = \rho M(w), \tag{4.1}$$

where ρ is defined in (1.3). In particular,

$$D(w) \leq \rho M(w). \tag{4.2}$$

Proof. Using (2.1) and (3.4),

$$\begin{aligned} D + T &= 2D - 2k^2 K + k^4 M \\ &= \frac{\beta}{2} Q - \frac{2\gamma}{3} S + k^4 M. \end{aligned}$$

Also, for $r = |w|^2$,

$$\frac{2\gamma}{3} \int r \left(r - \frac{3\beta}{8\gamma} \right)^2 \mathrm{d}x = \frac{2\gamma}{3} S - \frac{\beta}{2} Q + \frac{3\beta^2}{32\gamma} M.$$

Adding proves (4.1); all three terms on the left are nonnegative, giving (4.2). □

Lemma 4.2 (Fourier sup-norm estimate)**.** *For every $u \in H^2(\mathbb{R}^2;\mathbb{C})$ and $\eta > 0$,*

$$\|u\|_\infty^2 \le \frac{D(u) + \eta M(u)}{8\sqrt{\eta}}. \tag{4.3}$$

For every nonzero u, optimizing in η yields

$$\boxed{\|u\|_\infty^2 \le \frac{1}{4}\sqrt{M(u)D(u)}.} \tag{4.4}$$

Proof. Adopt the Fourier convention $\widehat{u}(\xi) = \int e^{-ix\cdot\xi} u(x)\,\mathrm{d}x$. Fourier inversion, weighted Cauchy–Schwarz, and Plancherel show

$$\begin{aligned} |u(x)|^2 &\le \frac{1}{(2\pi)^4}\left(\int_{\mathbb{R}^2} \frac{\mathrm{d}\xi}{|\xi|^4 + \eta}\right)\left(\int_{\mathbb{R}^2} (|\xi|^4 + \eta)\,|\widehat{u}(\xi)|^2\,\mathrm{d}\xi\right) \\ &= \left[\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{\mathrm{d}\xi}{|\xi|^4 + \eta}\right][D(u) + \eta M(u)]. \end{aligned}$$

Polar integration gives

$$\frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} \frac{\mathrm{d}\xi}{|\xi|^4 + \eta} = \frac{1}{2\pi}\int_0^\infty \frac{r\,\mathrm{d}r}{r^4 + \eta} = \frac{1}{8\sqrt{\eta}}.$$

This proves (4.3). For $u \neq 0$, both $M(u)$ and $D(u)$ are positive (a square-integrable harmonic function on $\mathbb{R}^2$ is zero). The choice $\eta = D(u)/M(u)$ gives (4.4). The estimate first follows for Schwartz functions and extends by density to H^2. □

Proposition 4.3 (Stationary sextic/quartic ratio)**.** *For every nonzero w satisfying (3.4),*

$$\boxed{\frac{S(w)}{Q(w)} \le \|w\|_\infty^2 \le \frac{\sigma}{4}M(w).} \tag{4.5}$$

Proof. By (4.4) and (4.2),

$$\|w\|_\infty^2 \le \frac{1}{4}\sqrt{M(w)D(w)} \le \frac{\sqrt{\rho}}{4}M(w).$$

Also $S(w) = \int |w|^2\,|w|^4\,\mathrm{d}x \le \|w\|_\infty^2\, Q(w)$; since $w \neq 0$, $Q(w) > 0$. □

5 A quantitative energy-per-mass gap

Proposition 5.1 (Strict decrease at the infimum)**.** *Let $0 < a < b$ and assume $a + b < 2m_B$. Set*

$$e(s) = \frac{I(s)}{s}, \qquad h = 1 - \frac{a+b}{2m_B} > 0.$$

Then

$$\boxed{e(a) - e(b) \ge \frac{(b-a)h}{a^2}[-I(a)] > 0.} \tag{5.1}$$

Proof. Choose any minimizing sequence (w_n) at mass a, adjusted as in Lemma 3.1. Set $t = b/a > 1$. Amplitude rescaling has mass $M(\sqrt{t}\, w_n) = ta = b$ and transforms the energy according to

$$\frac{H(\sqrt{t}\, w_n)}{t} - H(w_n) = (t-1)\left[-\frac{\beta}{2}Q(w_n) + \frac{\gamma(t+1)}{3}S(w_n)\right]. \tag{5.2}$$

By Proposition 4.3, using $M(w_n) = a$,

$$\begin{aligned} -\frac{\beta}{2}Q(w_n) + \frac{\gamma(t+1)}{3}S(w_n) &\le \left[-\frac{\beta}{2} + \frac{\gamma\sigma(t+1)a}{12}\right]Q(w_n) \\ &= -\frac{\beta}{2}\left(1 - \frac{a+b}{2m_B}\right)Q(w_n) \\ &= -\frac{\beta h}{2}Q(w_n). \end{aligned}$$

Since $T \ge 0$ and $S \ge 0$, we have

$$H(w_n) \ge -\frac{\beta}{2}Q(w_n).$$

Consequently,

$$\begin{aligned} \frac{I(b)}{t} &\le \frac{H(\sqrt{t}\, w_n)}{t} \\ &\le H(w_n) - (t-1)\frac{\beta h}{2}Q(w_n) \\ &\le [1 + (t-1)h]H(w_n). \end{aligned} \tag{5.3}$$

Pass to the limit $n \to \infty$ and divide by a:

$$e(b) \le [1 + (t-1)h]e(a).$$

It follows that

$$e(a) - e(b) \ge -(t-1)he(a) = \frac{(b-a)h}{a^2}[-I(a)].$$

The right side is strictly positive by Lemma 2.2. In particular the positive defect is independent of the minimizing-sequence index and cannot vanish upon taking the infimum. □

Remark 5.2 (The endpoint and larger-mass pairs)**.** *If $0 < a < b \le m_B$, then $a + b < 2m_B$, including when $b = m_B$. The argument also applies to some comparisons with $b > m_B$, provided $a + b < 2m_B$; it does not assert monotonicity for arbitrary masses.*

6 Strict binding and the frozen-parameter corollary

Proof of Theorem 1.1. Fix $0 < m \le m_B$ and $0 < \mu < m$. Since

$$\mu + m < 2m \le 2m_B, \qquad (m-\mu) + m < 2m \le 2m_B,$$

Proposition 5.1, applied to (μ, m) and $(m-\mu, m)$, yields

$$e(\mu) > e(m), \qquad e(m-\mu) > e(m).$$

Both μ and $m - \mu$ are strictly positive; multiplying the two inequalities by those masses and adding gives

$$\begin{aligned} I(\mu) + I(m-\mu) &= \mu e(\mu) + (m-\mu)e(m-\mu) \\ &> [\mu + (m-\mu)]e(m) = I(m). \end{aligned}$$

This is (1.4) for every interior split. □

Corollary 6.1 (Exact frozen-parameter check)**.** *Consider $k = 1$, $\beta = \gamma = b_0$, and $m = m_0$, where the parameters are the exactly represented binary64 values*

$$b_0 = \texttt{0x1.05f1d178c75ebp+2}, \qquad m_0 = \texttt{0x1.45b8a0bad2714p+1}.$$

Viewed as rational numbers, they equal

$$b_0 = \frac{4608178181141995}{1125899906842624}, \qquad m_0 = \frac{1432536998431173}{562949953421312}.$$

For these coefficients, $m_0 < m_B$. Therefore the strict-binding inequality holds at mass m_0 for all $0 < \mu < m_0$.

Proof. Because $k = 1$ and $\beta = \gamma = b_0$,

$$\rho = 1 + \frac{3b_0}{32} = \frac{49853331562389953}{36028797018963968}, \qquad m_B = \frac{3}{\sqrt{\rho}}.$$

Exact rational arithmetic (before rounding any square roots) gives

$$9 - \rho m_0^2 = \frac{45470871232408758421247666267839197890077559 1}{11417981541647679048466287755595961091061972992} > 0. \tag{6.1}$$

As all quantities are positive, this proves $m_0 < 3/\sqrt{\rho} = m_B$. For orientation only, the decimal values are

$$m_0 = 2.54469689485712\ldots, \qquad m_B = 2.550345636351549\ldots.$$

Apply Theorem 1.1. □

Remark 6.2 (Scope of the frozen corollary)**.** *The notation "D4" used in the associated computational project refers there to a particular square-symmetric configuration in two spatial dimensions. Corollary 6.1 is a statement about an unrestricted whole-space functional on $\mathbb{R}^2$; it does not establish identification with a computed periodic branch, uniqueness of minimizers, spectral stability of a branch, or a lifetime classification for four spatial dimensions.*

7 Discussion and limitations

The proof uses two complementary scalings. The mass-preserving spatial dilation $u \mapsto \lambda u(\lambda \cdot)$ supplies an exact identity on adjusted minimizing sequences. The amplitude scaling $u \mapsto \sqrt{t}\, u$ then compares two masses. The positive sextic amplitude contribution prevents a universal comparison for *all* profiles; the stationary ratio estimate (4.5) controls that contribution on the adjusted sequences. In particular, negativity of $I(m)$ alone would not suffice to justify the comparison.

The threshold m_B is a *sufficient* mass bound, not a proved sharp boundary of strict binding. This paper does not claim equality, failure of binding, or optimality of its constants for masses above m_B. Likewise, strict binding is distinct from attainment and compactness: concentration–compactness can use strict subadditivity to exclude certain splitting scenarios, but any existence or dynamical stability theorem requires additional hypotheses and arguments.

For completeness, ordinary subadditivity $I(a + b) \le I(a) + I(b)$ follows by approximating mass-a and mass-b near-minimizers by compactly supported smooth profiles, restoring their exact masses by a factor tending to one, and translating one profile until the supports are separated. The mass, quadratic derivative form, and local nonlinear terms then add exactly; approximation and continuity of H allow passage to the infima. The strict conclusion of Theorem 1.1 is not a consequence of that elementary construction, but of Proposition 5.1.

Reproducibility. The associated development notes, including independent dilation and arithmetic audits, are available in the public repository at

`https://github.com/rickyjreyes/geometry_of_resonance/tree/main/docs/d4_binding`.

The theorem and its proof above are self-contained and do not depend on numerical evidence or external computational artifacts.

References

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